Variant Selection Rules Govern Gene Level Validation of Single Cell Foundation Model Signatures

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Abstract

Pipelines that check genes ranked by single cell foundation models against protein damaging mutations score individual amino acid substitutions but report verdicts for whole genes, so a selection rule must decide which substitution represents each gene. Despite this, the specific rules used are scarcely reported. In our experiment, we ranked genes with a frozen scGPT backbone and a linear classifier trained on four Neftel glioblastoma cell states, which contained 6,57 malignant cells and 27 patients, then confirmed candidates against mutations from 558 TCGA glioblastoma patients using pathogenicity values per patient from AlphaMissense at the acclaimed 0.564 threshold value. Keeping all other factors constant, we compared four rules that intake the most recurrent variant, the highest or lowest scoring variant, or the variant that is closest to the start of the protein. In combination, these rules confirm between 169-437 genes, and 40.1% of the genes confirmed under the pre registered rule have a dependence on the specific choice. Their difference between rules withholds fifteen varied thresholds, spanning from 1.89-3.01 fold, and on 961 genes that are shared between two different diseases (lung adenocarcinoma and melanoma), it rises with the share of multi-variant genes from 2.79 to 9.19 and 32.72 fold, where the rule dependent share of the default set reaches 72-90%. These differences are confined to genes whose variants fall on both sides of the threshold, and lists are nominated without any foundation model behaving in any type of replicated behavior. At a fixed panel of 120 genes, every rule's genes beat random genes at classifying the four states, yet the rules differ at the same amount as two random draws from a singular set, and a prespecified check on four known genes passes or fails depending on the rule. Overall, we found that a confirmed gene list cannot be interpreted without its specific variant rule.

1 Introduction

Two single cell foundation models of scGPT and Geneformer are pretrained on millions of cells and then used to classify cell states and rank genes by their significance to a classification [7, 23]. In cancer terms, these rankings are read as lists of candidate driver genes. Hence, Glioblastoma is used as a common test case since its malignant cells fall into four transcriptional states: astrocyte (AC), mesenchymal (MES), neural progenitor (NPC) and oligodendrocyte (OPC) along with moving between them [17]. A specific gene which can seperate these states can drive them, respond to them, reflect one patient or be noise, and expression data alone cannot tell these cases apart. BEcause the attention weights of these models capture structure without improving prediction over simpler gene level methods [14], utilizing a second check that doesn't depend on expression is a suitable choice.

One choice is through protein level evidence. AlphaFold provides structure confidence for nearly ever human protein [13, 24] and AlphaMissense provides all possible single amino acid substituion a pathogenicity score from 0-1 [6]. A validator can keep a ranked gene only if it carries a substitution predicted to be damaging. We call the genes that pass the confirmed set.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

The innate difficulty is a mismatch of units. The model ranks genes, the evidence score substitutions, and a gene in a patient cohort usually carries several substitutions with different scores. Specifically, TP53 in our glioblastoma cohort carried 62 distinct missense variants, which scored from 0.084-1.000. In order to provide TP53 with a verdict, the validator needs to choose which score represents it, the most frequent variant, the highest score, the lowest score, or utilize a different choice. We classified this as the *selection rule*. In the pipelines, we understand it as an implementation detail that is not disclosed, never varied and invisble in the output, which only contains a list of genes.

We treated the selection rule as the object of study and explained each result from the structure of the variant data. There exists four rules which produced confirmed sets that differ by around 2.59 fold at the disclosed threshold; the difference extrapolates as the threshold becomes stricter and reaches 32.79 fold in melanoma on genes shared with glioblastoma; it barely changes classification performance, and a check on genes with known answers passes or fails depending on the rule.

2 Related Work

scGPT [7], Geneformer [23] and CellFM [26] are transformer models pretrained on large collections of single cell expression profiles. Their evaluations usually report classification accuracy instead of asking how gene lists drawn from them depend on later processing. On the protein side, AlphaFold and its database provide per residue structure confidence for the human proteome [13, 24], AlphaMissense scores every possible missense substitution and sets 0.564 as the threshold for likely pathogenic [6], and ESM1b gives a comparable protein language model score [2]. These scores approximate damage to protein function, not measured destabilisation of the fold, and we make no claim on folding. ClinVar [16] and MaveDB [8] provide independent tests, but every one of these resources scores variants and none of them say how to to turn several different variant scores into one gene verdict. Other fields aggregate variants by design. Two of these methods include utilizing rare variant burden tests to indicate collapsing on variants within a gene[**?**], and dNdScv calls drivers from a gene's excess non synonymous mutations [**?**]. In both of these methods, the aggregation is included inside the accepted methodology, which differs from most validator pipelines as these pipelines refer to utilizing this aggregation in the implementation instead. This step is the analytic choice studied in work on reproducibility. Choices that are made after visualizing data changes conclusions even without deliberate searching [11]. Additionally, multiverse analysis runs every reasonable version of a static analysis and reports the range of results [22]. On top of this, vibration of effects measures how associations shift across model specifications [19], and many analyst studies found that independent teams reach varying conclusions from a single dataset [21, 1]. Conversely, we varied a single choice and identified in advance the genes it could possibly change.

3 Methods

3.1 Cohorts and Preprocessing

The single cell Neftel et al [17] glioblastoma cohort contains 6,576 malignant cells from 27 different patients across 17,268 unique genes, which were drawn after careful quality control removed 1,354 of the original 7,930 cells. In this process, cells were retained if they expressed 200 or more different genes. On the other hand, genes were retained if they were expressed by 3 or more unique cells. Furthermore, both of these counts were scaled to 10,000 per cell and trasnformed by the function $\log(1 + x)$ by utilizing scanpy [25]. Instead of individual cells, patients were split into individual training, validation, and test cohorts, by a split ratio of 15:6:6. Additionally, the source code checks to ensure no single patient appears in two sets (duplication)(Figure 1). All mutation and copy number data are collected from the TCGA PanCancer Atlas studies that are distributed throughout the cBioPortal platform [5, 10, 12] for glioblastoma, which contained 558 patients and 68,902 individual mutation records. A similar process was performed for cutaneous melanoma [3] and lung adenocarcinoma [4]. Both of these diseases differ from glioblastoma since it is heavily affected from copy number changes to contain a variety of unique point mutations. Each of these cohorts passed four integrity checks, which included realistic patient identifiers, genuine missing clinical values, several collection sites, and expected mutation rates in known drivers.

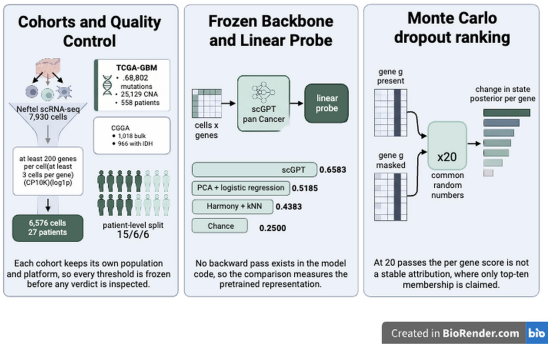


Figure 1: From cells to candidate genes: cohorts under frozen quality control and a patient level split, a frozen scGPT backbone with a linear probe, and Monte Carlo dropout gene ranking.

3.2 Backbone and Gene Ranking

Utilizing the pretrained scGPT pancancer checkpoint [7] without updating its weights. We utilized a logistic regression component that was trained as a classifier [20] on balanced class weights, and fitted on the training patients to predict the four cell states from the frozen model's cell embeddings. Individual genes were ranked through a Monte Carlo dropout [9] from taking the average change in predicted state probability when the gene is removed, and was performed on 20 paired passes. Specific intervals arise from resampling patients or gene panels. For a second backbone we utilized a frozen version of Geneformer [23], which ran on the exact same specification regarding classifiers, patient split and gene panels as scGPT.

3.3 Structural Validator and Selection Rules

For each individual gene, the validator joins the cohort's somatic mutation records to AlphaFolds confidence value (pLDDT) and AlphaMissense's pathogenicity value, which were both released under CC-BY-4.0 [24, 6]. In our case, a missense variant counts as damaging if its AlphaMissense score is 0.564 or above, which is AlphaMissense's published likely pathogenic threshold. Each gene ultimately results in one of five different outcomes as seen in Figure 2. Unfortunately, specific genes needing protein stability validation estimates are unable to be processed since they aren't available.

Any gene with several scored variants is not assigned to a score until a selection rule selects one. We perform this experiment with four rules: R1 uses the variant observed in most patients, R2 utilizes the highest scoring variant, R3 uses the lowest scoring variant, and R4 uses the variant that is closest to the beginning of the protein. On top of this, R5 is an equivalent classification in terms of ruling as compared to R2 since it confirms if any variant passes, and R6 is equivalent to R3 in terms of ruling since it confirms only if every single variant passes. Additionally, every gene confirmed by R3 is also confirmed by R2, so we summarized the rules by their fold differences (calculated by the size of the R2 set / the size of the R3 gene set).

A confirmed gene is eligible for a model if it also appears in the Neftel dataset cohort and the scGPT bandwidth of vocabulary, which allows them to enter a classification panel. All and eligible counts of genes for each rule are as follows: 282 and 250 for R1, 437 and 395 for R2, 169 and 152 for R3, and 278 and 246 for R4.

We disclaim on the fact that noting on any of the datasets that we utilized in our experiment were registered externally on our source code, which is clearly outlined in our codebases version history. Our initial running on August 10th 2026 included the output of the four gate genes and their expected outcomes, precluding the first recorded validator output which was logged on the 20th of August. On September 6th, we utilized AlphaMissense to replace the EMS1b score that was unobtainable by us, and adopted its published 0.564 threshold on that same day; though, our configuration logs detail the fact that no other value was tried. R1 is the rule the validator was written with, but it entered our codebase on September 10th in the same commit as the first comparison of all four rules, and the codebase defines R2 and R4 wasn't apparent in our codebase. As a result, we are unable to prove that R1 was fixed before any preceding results, and we treat the four rules as equal alternatives rather than as a primary rule and its sensitivity checks. The alternatives were not freely chosen., since R2 and R3

bound every rule that keeps a single score variant per gene (Section 4.2). Furthermore, the threshold sweep, the cross cancer comparison and the fixed size comparison followed on the 19th and 20th of September, and the frozen gene list and shred universe on the 30th of September.

3.4 Sensitivity Analyses

Figure **??** summarises four of the analyses below. We repeated the validator at fifteen thresholds from 0.20 to 0.90 in increments of 0.05. For the comparison across cancers, we fixed the 1,082 glioblastoma genes with a scored variant before looking at the other cancers and applied the same list to them, treating a gene without a scored variant as missing rather than negative. The 961 genes scored in all three cohorts form the main comparison. A gene's verdict is dependent on the rule exactly when its lowest variant scores are below the threshold and when it is highest above the threshold, and we call these specific genes as *straddling*. To predict them, we fitted a regularised logistic regression on 649 genes and tested it on other 433 genes, where the stratified split was 60/40, and we set our seed value to 1729, using eight properites of each gene's variants, as seen in Figure 5, and excluding the straddling indicator, which defines the outcome. Larger panels are able to classify more accurately, so in order to separate rules from size classification we drew 200 random panels of 120 genes from each rule's model eligible set (the smallest set has 152 genes) and 200 panels of 120 genes from the 1,171 eligible genes outside every confirmed set as a baseline. Rules were compared with a Kurskal-Wallis test and pariwise Mann Whitney tests with Holm correction. To cover rules between the extremes, we represented each gene by the qth quantile of its scores for q from 0-1 in steps of 0.05, and by one of

4 Results

4.1 The Rule Dependence of Confirmed Sets

At the 0.564 threshold, R2 confirms 437 genes, R1 confirms 282 genes, R4 confirms 278 genes, and R3 confirms 169 genes. The largest set is 2.59x the smallest. R1 and R4 agree closely because of the Jaccard Overlap, and they both share genes over genes in either set (0.979). Of the 1,082 genes with a scored variant, 594 carry more than one variant, and their scores within a gene span a median range of 0.342 in the 0-1 scale. Of those 594 genes, 268 are straddling, with at least one variant on each side of the threshold. Only these genes can change the verdict, and R2 and R3 disagree on all of them (437-169=268). We use this equality as a check on the implementation, not as a reported finding. Of the 282 genes that R1 confirms, 169 are confirmed under each rule, 130 have a single variant and 39 have all variants above the threshold. The specified 169 genes are exactly the R3 set. The remaining 113 genes, 40.1% of the R1 set, depend on that rule.

Every confirmed set in this cohort is therefore the same 169 genes along with a strata of the 268 straddling genes, and the rules differ only in which portion they add. Each rule answers a different question. R2 asks whether any observed changed to the gene is damaging, that is, whether the gene can be damaged. R3 asks whether every observed change is damaging. R1 asks whether the change that tumors carry most often is damaging. Recurrence across patients is the usual evidence that a mutation was selected during tumor growth, and R1 is the only rule that utilizes it. R4 agrees with with R1 in almost every gene, which indicates that here the variant nearest to the start of a protein is usually also the most recurrent. Straddling genes make up 45.1% of the genes with several variants,

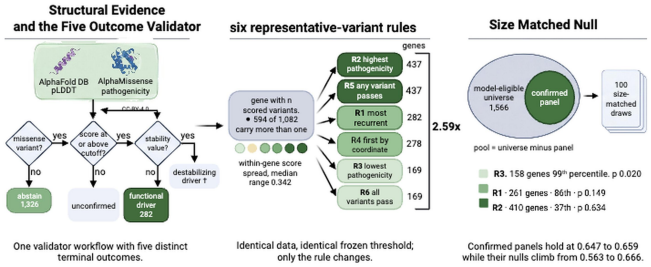


Figure 2: Provides per set evidence resolves to one of five outcomes, two of them inactive in this run. A selection rule then decides which scored variant represents each gene, and the rules yield different confirmed sets from identical data.

and 49.8% (243 of 488) of those than can enter a classification panel, so the genes a downstream would actually use are slightly more prone to the rule than the whole cohort.

4.2 Threshold Sensitivity

The fold difference between R2 and R3 rises from 1.89 at a threshold of 0.20 to 3.01 at 0.90, and it is above 2 at every threshold from 0.25 and above (Figure 3). R1 and R4 remain close to each otehr at every threshold.

The rise has a simple explanation. Consider a gene with k variants, each of which passes the threshold with a probability p. R2 confirms the genes if at least one variant passes, which happens with probability $1-(1-p)^k$. R3 confirms it only if all variants pass, which happens with the probability p^k. A stricter threshold lowers the probability p. For $k > 1$, lowering p shrinks p^k much faster than $1-(1-p)^k$, so the ratio grows, and at a leinient threshold both approach towards 1. The observed cunts follow this pattern. Between thresholds of 0.20-0.90, R2 keeps 31% of its genes (226 of 740) while R3 keeps 19% of its genes (75 of 392). A stricter threshold, often chosen for safety reasons, therrfore makes the result more dependent on the rule instead of being less dependent.

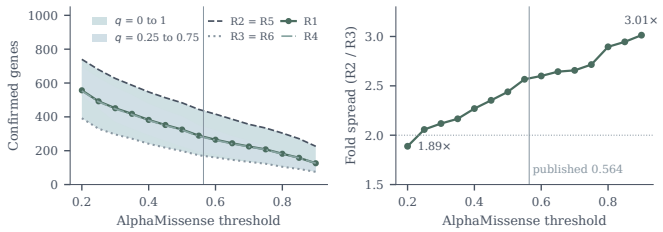


Figure 3: Threshold sweep on glioblastoma. Left, confirmed genes per rule at fifteen AlphaMissense thresholds; R5 and R6 coincide with R2 and R3. Right, fold difference between R2 and R3; the vertical line marks the published 0.564 boundary and the dotted line marks 2.

4.3 Cross Cancer Replication

On the 961 genes scored in all three cohorts, the rules differ by 2.79 fold in glioblastoma, 9.19 fold in lung adenocarcinoma and 32.79 fold in melanoma (Figure 4, Table 1). The order matches the fraction of genes carrying more than one variant (0.586, 0.850, and 0.967) and the median score range within those genes (0.344, 0.722, and 0.818).

The same probability argument explains this pattern. Melanoma and lung adenocarcinoma carry many more somatic mutations than glioblastoma because of ultraviolet light and tobacco exposure [3, 4]. Most are passengers, which do not drive the tumor and spread across the score range. A randomly chosen substitution in the glioblastoma candidate proteins scores above 0.564 about 43% of the time, so a gene with three random variants would pass R2 with a probability of 0.82 and R3 with a probability of 0.08. With more variants per gene, R2 confirms almost every gene, R2 with a probability of 0.08. With more variants per gene, R2 confirms almost every gene and R3 confirms almost none. R2 rises from 394 to 787 genes and R3 falls from 141 to 24. R1 and R4 each look at a single variant, so their counts change slightly (R1 from 248 to 260, and R4 from 209 to 259). R1 confirms between 25.8% and 27.1% of the 961 genes in all three cancer. A rule that reads one variant per gene should confirm roughly the share of observed variants that clear the threshold, which is 28.6% in glioblastoma, and R1 satisfies this. It's output follows how damaging the variants tumors carry are, not how many there are. As passengers accumulate, however, the variant nearest to the start of the protein is more likely to be a passenger, while the most recurrent variant is still one that tumors select. Under R2 and R3, confirmation therefore measures how often a gene is mutated as well as how damaging its mutations are, and gene lists from cancers with different mutation rates cannot be compared directly.

4.4 Predictors of Rule Dependence

On the 433 test genes, the model classifies 426 correctly (98.4%), with 105 true positive, 321 true negatives, 5 false positives and 2 false negatives, and an AUROC of 0.998. This accuracy is close to

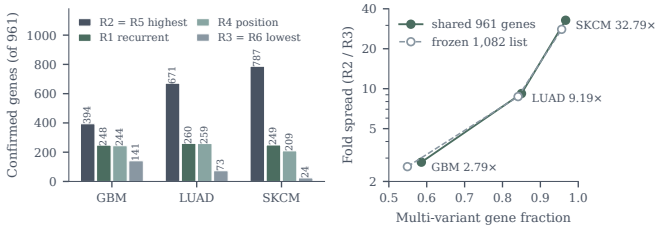


Figure 4: Cross cancer replication. Left, confirmed genes per rule on the 961 genes scored in all three cohorts. Right, fold difference between R2 and R3 against multi variant gene fraction, on the shared 961 genes (filled) and the frozen 1,082 gene list (open); log scale.

Table 1: Rules on the 961 genes scored in all three cohorts at the 0.564 threshold. Straddle counts genes with variants on both sides of the threshold and equals R2 minus R3 in every cohort. The last column applies the frozen 1,082 gene list without intersection (982 and 1,049 genes scored in lung and melanoma).

Cohort	Multi frac.	Med. range	Straddle	R1	R2	R3	R4	Fold	Fold, 1,082
Glioblastoma	0.586	0.344	253	248	394	141	244	2.79	2.59
Lung adenocarcinoma	0.850	0.722	598	260	671	73	259	9.19	8.72
Melanoma	0.967	0.818	763	249	787	24	209	32.79	28.00

guaranteed, because two of the eight inputs, the fraction of variants above the threshold and their median distance from it, use the threshold that defines the outcome. It shows only that the outcome is recoverable for unseen genes. The more informative result is the coefficient for the range of of scores within a gene, which doesn't use the threshold and has the largest effect (7.10, $p = 0.14$, Figure 5a).

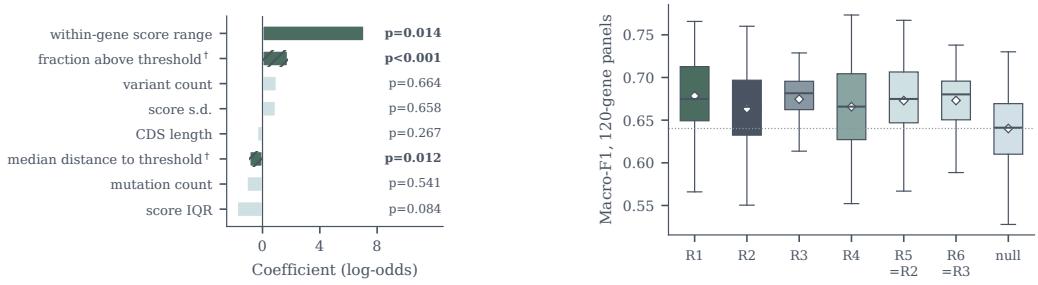
Whether a gene depends on the rule is set by how much its variants disagree, not by how often it is mutated. IDH1 has three variants and does not depend not the rule because all three have scores above 0.98. TP53 depends on the rule because its 62 variants cover almost the whole scale. The number of variants add nothing once the range is known. Wide spread is the unusual case rather than the exception. Half of the 594 genes with several variants span more than 1/3 of the score scale (median range of 0.342), so one benign and one damaging variant in the same gene si common. A pipeline can therefore measure its sensitivity to the rule from the mutation table alone, before any model is run. The number of candidates whose score range crosses the threshold is the largest number of genes the rule can change.

4.5 Classification at Fixed Panel Size

At a panel size of 120 genes, every rule's confirmed genes classify the four states better than the baseline panels (Figure 5b). Mean macro-F1, the average of the F1 scores for the four states, is 0.678 for R1, 0.675 for R3, 0.666 for R4 and 0.663 fro R2, against 0.640 for the baseline. Holm correct p values against the baseline range from 1.6×10^{-5} to 3.8×10^{-15}. Across all six named rules the Kruskal-Wallis test gives $p = 0.016$, but only the comparison of R1 with R2 remains significant after correction ($p = 0.044$). R2 and R5 draw from the same gene set with independent random draws and still differ by 0.0095, close to the largest difference between distinct rules, 0.015.

This design also tests directly whether the straddling genes carry information about cell state. The model eligible R3 set is the 152 genes confirmed under every run, while the R2 set adds 243 eligible straddling genes, so a random 120 gene panel from R2 is about 60% straddling genes and a panel from R3 contains none. If the straddling genes carried state information, R2 panels would score higher. They score slightly lower (0.663 against 0.675). The genes that the rules disagree about therefore add not no state information beyond the genes confirmed under every rule. This is why classification performance cannot reveal which rule produced a gene list. It leaves open why every rule outperforms the baseline. The original 32 gene validator panel suggests an answer. Its genes have a mean training set expression of 0.560 against 0.267 for the eligible pool, and it score is around 0.461, below all 20 random panels matched on mean expression and expression variance (mean value of 0.481). Genes that reach a confirmed set tend to be highly expressed, and highly expressed genes separate states better, so the advantage of confirmed panels over random ones may reflect expression

level rather than mutation evidence. Whether the 120 gene comparison survives expression matching has not been tested.



(a) (b)

Figure 5: (a) Coefficients of the instability model on glioblastoma training genes, straddle indicator excluded; dark bars mark $p < 0.05$, and hatched features (†) are computed from the threshold that defines the label. (b) Macro-F1 of 200 random 120 gene panels from each rule's model eligible confirmed set and of 200 panels from eligible genes outside every confirmed set (null, dotted line at its mean); diamonds mark means. R5 and R6 resample the R2 and R3 sets and show how far identical sets differ by chance.

4.6 Pre Registered Gate Genes

Before the study, we specified a four gene check. TP53 and IDH1 should be confirmed, and EGFR and RPRM, which are altered mainly be amplification and silencing, should not. IDH1 has three variants scoring between 0.987 and 0.998 and is confirmed under all four rules. RPRM has one variant scoring near 0.341 and is confirmed under none. TP53 has 62 variants scoring from 0.084-1.000 and is confirmed under three of four rules. EGFR has 51 variants scoring from 0.061 to 0.999 and is confirmed under two variants (Table 2).

Table 2: Gate genes under each distinct rule ($Yes = confirmed$); the gate expects TP53 and IDH1 only.

Gene	Var.	Range	R1	R2	R3	R4	Class
IDH1	3	0.987–0.998	Yes	Yes	Yes	Yes	independent
TP53	62	0.084–1.000	Yes	Yes		Yes	dependent
EGFR	51	0.061–0.999	Yes	Yes			dependent
RPRM	1	0.341					independent

The probability argument predicts which genes are stable. Those with few variants or with variants of similar effect. IDH1 is stable because it is mutated almost only at one position, R132, and RPRM because it has one observed variant. TP53 and EGFR are among the most frequently mutated genes in glioblastoma, so the same evidence that makes them obvious test cases also makes their verdicts depend on the rule. Test genes should therefore be chosen for a narrow score range rather than for how well known they are, or the check should be repeated under every rule.

4.7 Baselines and Backbone Robustness

On the 91 gene panel used for classification, the frozen scGPT model with a logistic regression classifier reaches a macro-F1 of 0.658 averaged over three different seeds [0.607, 0.697]. Principal component analysis with logistic regression reaches 0.519 [0.442, 0.555], and Harmony [15] with k nearest neighbours reaches 0.438 [0.385, 0.460] (Table 3). The pathogenicity scores also agree with independent labels. AlphaMissense separates pathogenic from benign ClinVar variants with an AUROC of 0.963 [0.952, 0.972] across 1,106 variants, and it predicts loss of function measured experimentally in 7,448 TP53 variants with 0.807 [0.796, 0.818]. With patients correctly linked across data types, 282 of 2,503 candidate genes (11.27%) can be confirmed under R1. Accurate variant scores do not guarantee consistent gene verdicts. The rules disagree on 268 genes not due to any score being objectively wrong but because they rest the verdict on different valid scored variants.

Table 3: Macro-F1 on Neftel test patients, 91 gene panel; patient clustered bootstrap 95% CI.

Method	F1	95% CI
scGPT, frozen	0.658	[0.607, 0.697]
PCA and log. reg.	0.519	[0.442, 0.555]
Harmony and kNN	0.438	[0.385, 0.460]

With the same 91 gene panel, classifier, patient split and seed, Geneformer reaches a macro-F1 of 0.660 [0.552, 0.706] and scGPT 0.654 [0.607, 0.697] (Figure 6). Geneformer's interval rests on five patients. scGPT is more accurate on AC cells (F1 scores 0.87 against 0.71) and Geneformer's on OPC cells (0.65 against 0.34), and the two models share only 10 of their 40 top 10 genes across the four states. On the independent CARE cohort [18], the authors' published state programs reproduce their own four state labels with a cell level macro-F1 of 0.985 and a mean per sample macro-F1 of 0.862 (95% CI 0.823 to 0.895) across 120 samples and 132,493 malignant cells. This checks the labels, not either backbone. OPC is the least frequent state among the cells used to rank genes (555 of 3,791), and it is where the two backbone differ most, which suggests that the choice of backbone matters most for rare states. A macro-F1 score summaries predictions, not the genes behind them two models within 0.006 share 1/4 of their top genes, so the list depends on the nominating model than on the rule.

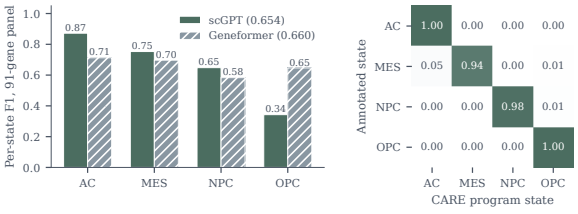


Figure 6: Left, per state F1 of scGPT and Geneformer on the Neftel test patients with the 91 gene panel (overall macro-F1 in the legend). Right, row normalised confusion matrix of public CARE state programs against author annotations across 120 samples.

5 Discussion, Limitations and Future Work

Each component that is usually checked passes in this study. The backbone beats standard methods, the variant scores agree with independent labels, and the states reproduce in another cohort. The confirmed list still changes by up to 2.59 fold in glioblastoma and 32.79 fold in melanoma at a step none of these checks examines, where several variant scores become one gene verdict. The change is confined to straddling genes, whose number depends on how many variants each gene carries and how widely their scores spread, which are both measurable before any model is trained. The probability argument of Section 4.2 is not specific to glioblastoma, single cell models or AlphaMissense, and any study drawing gene conclusions from several variant scores can bound the effect in advance by its own straddle fraction. The rules suit different purposes. R2 suits a screen that must not miss a damaged gene, but with many passengers it still confirms most genes. R1 asks whether the mutation tumors select is damaging and is the only rule both unaffected by passenger count and tied to what a driver call is meant to establish, so we consider it the most defensible default. The usual checks miss the choice because well known test genes are the most rule dependent (Section 4.6) and straddling genes add no state information (Section 4.5). Unlike most multiverse studies [22, 19], the range here is predictable from the data, so we recommend reporting the rule, the straddle fraction and the genes confirmed under every rule, and comparing cancers only with a recurrence based rule or that shared set.

Two inputs of the rule dependence model use the threshold that defines the outcome so its test accuracy shows consistency, not predictive skill. A model on threshold free inputs, tested at other thresholds and in other cancers, would settle this. Three cohorts cannot separate mutation rate from other differences between cancers. The rules are discrete points, and the qth quantile of a gene's scores runs from R3 ($q = 0$ to R2 ($q = 1$) and would map the full sensitivity. The stability branch never ran for the lack of stability estimates, and AlphaMissense replaced the ESM1b score in our protocol. A third, independent source of evidence may narrow the difference between rules. The backbone comparison uses one seed and six test patients. CellFM failed to run in our experiment, and the fixed size results use only Neftel labels, so repeating them on the CARE cohort would test whether they can generalize.

6 Conclusion

Gene level validation from variant evidence requires choosing which variant represents each gene, and this choice is rarely reported. With the cohort, scores and threshold fixed, four reasonable rules confirm under the pre registered rule dependent on it. It stays stays between 1.89-3.01 fold across fifteen thresholds and reaches 9.19 fold in lung adenocarcinoma and 32.79 fold in melanoma, following how many genes carry several different variants. It is confined to genes whose variants straddle the threshold, so its maximum size can be read from the mutation table before any model is trained. Neither classification performance nor a check on well known genes reveals which rule was used. A confirmed gene list should be reported with its rule, its straddle fraction and its rule independent genes.

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